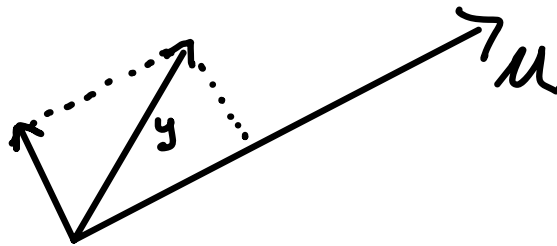


Class 33: Orthogonal Projections

We saw last class that if $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ is an orthogonal basis for some subspace W in $\mathbf{R}^n$, then any vector $\mathbf{w}$ in W may be written as $c_1 \mathbf{u}_1 + c_2 \mathbf{u}_2 + \dots + c_k \mathbf{u}_k$ where each weight is computed by

Now if $\mathbf{y}$ and $\mathbf{u}$ are any vectors in $\mathbf{R}^n$ we can decompose $\mathbf{y}$ into the sum of a component along the direction of $\mathbf{u}$ and a component in a direction orthogonal to $\mathbf{u}$.



Geometrically $\hat{\mathbf{y}}$ is the point on the line spanned by $\mathbf{u}$ that is closest to $\mathbf{y}$.

We want to generalize this idea by trying to solve the problem of how to find the point in a nonzero subspace W that is closest to $\mathbf{y}$. We will call this point the projection of $\mathbf{y}$ onto W .

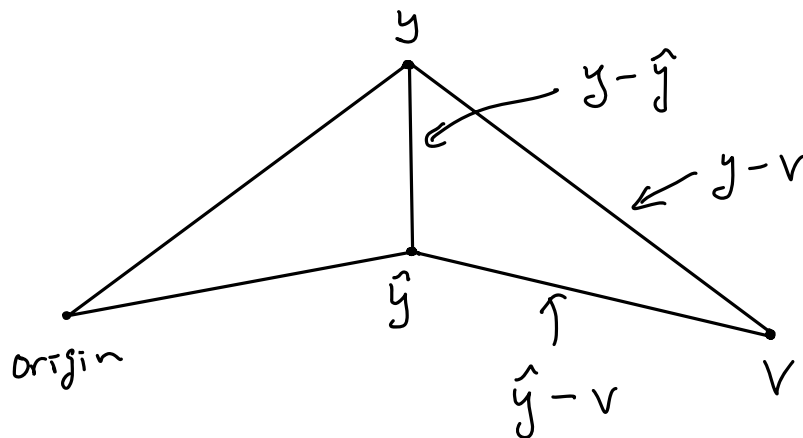
Let $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ be any orthogonal basis for a nonzero subspace W in $\mathbf{R}^n$

$$\text{Proj}_W \mathbf{y} = \hat{\mathbf{y}} =$$

$$\text{Let } \mathbf{z} = \mathbf{y} - \hat{\mathbf{y}}$$

Show that $\mathbf{z}$ is living in $W^\perp$ by showing that the dot product of $\mathbf{z}$ and each of the basis vectors for W is 0.

Key Idea: $\hat{\mathbf{y}} = \text{proj}_W \mathbf{y}$ is the vector in W that is closest to $\mathbf{y}$.



$\hat{\mathbf{y}}$ is in W and $\mathbf{v}$ is any other vector in W . We need to show that the distance from $\mathbf{y}$ to $\hat{\mathbf{y}}$ is less than the distance from $\mathbf{y}$ to $\mathbf{v}$.

Practice Problems:

Find the orthogonal projection of $\mathbf{y} = \begin{bmatrix} 6 \\ 3 \\ -2 \end{bmatrix}$ onto the space spanned by $\mathbf{u}_1 = \begin{bmatrix} 3 \\ 4 \\ 0 \end{bmatrix}$ and $\mathbf{u}_2 = \begin{bmatrix} -4 \\ 3 \\ 0 \end{bmatrix}$

Find $\hat{\mathbf{y}}$ for the same $\mathbf{y}$ using the vectors $\mathbf{e}_1$ and $\mathbf{e}_2$ instead of $\mathbf{u}_1$ and $\mathbf{u}_2$.

Find the shortest distance from $\mathbf{y}$ to the subspace $W = \text{span}\{\mathbf{u}, \mathbf{v}\}$ in $\mathbf{R}^4$ if

$$\mathbf{y} = \begin{bmatrix} 3 \\ -1 \\ 1 \\ 13 \end{bmatrix}, \mathbf{u} = \begin{bmatrix} 1 \\ -2 \\ -1 \\ 2 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} -4 \\ 1 \\ 0 \\ 3 \end{bmatrix}$$

